

Rearrangement Inequality

The **Rearrangement Inequality** states that, if $A = \{a_1, a_2, \dots, a_n\}$ is a permutation of a finite set (in fact, multiset) of real numbers and $B = \{b_1, b_2, \dots, b_n\}$ is a permutation of another finite set of real numbers, the quantity $a_1b_1 + a_2b_2 + \dots + a_nb_n$ is maximized when A and B are similarly sorted (that is, if a_k is greater than or equal to exactly i of the other members of A , then b_k is also greater than or equal to exactly i of the other members of B). Conversely, $a_1b_1 + a_2b_2 + \dots + a_nb_n$ is minimized when A and B are oppositely sorted (that is, if a_k is less than or equal to exactly i of the other members of A , then b_k is also greater than or equal to exactly i of the other members B).